Homework 5

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Problem 1

```
##                                                        Stratified by corticobl
##                                                         0
##   n                                                       495
##   trial (%)
##      Trial A                                              104 (21.0)
##      Trial B                                               60 (12.1)
##      Trial C                                               62 (12.5)
##      Trial D                                              216 (43.6)
##      Trial E                                               53 (10.7)
##   Randomized Group (Active = 1, Placebo = 0) = Yes (%)   281 (56.8)
##   Gender = Female (%)                                     220 (44.4)
##   Age (years) (mean (SD))                               56.92 (15.48)
##   History of Asthma = Yes (%)                              51 (10.3)
##   History of COPD = Yes (%)                                25 ( 5.1)
##   History of Diabetes = Yes (%)                           115 (23.2)
##   History of CHF = Yes (%)                                 20 ( 4.0)
##   History of Hypertension = Yes (%)                       213 (43.0)
##   History of Malignancy = Yes (%)                          24 ( 4.8)
##   History of Renal Impairment = Yes (%)                    44 ( 8.9)
##   Immunosuppressed Status = Yes (%)                        49 ( 9.9)
##   Body Mass Index (BMI) (mean (SD))                     30.63 (7.49)
##   NEWS Score at Baseline (mean (SD))                     3.39 (2.34)
##   Baseline Creatinine (mg/dL) (mean (SD))                1.14 (1.23)
##   Baseline C-Reactive Protein (CRP) (mean (SD))         82.08 (71.72)
##   Baseline Lymphocyte Count (mean (SD))                  1.15 (1.66)
##   Pulmonary Ordinal Scale (mean (SD))                    2.64 (0.74)
##   Viral Load Result (Log10) (mean (SD))                  4.32 (1.83)
##                                                        Stratified by corticobl
##                                                         1
##   n                                                       877
##   trial (%)
##      Trial A                                               71 ( 8.1)
##      Trial B                                               95 (10.8)
##      Trial C                                               90 (10.3)
##      Trial D                                              496 (56.6)
##      Trial E                                              125 (14.3)
##   Randomized Group (Active = 1, Placebo = 0) = Yes (%)   507 (57.8)
##   Gender = Female (%)                                     363 (41.4)
##   Age (years) (mean (SD))                               55.80 (14.71)
##   History of Asthma = Yes (%)                              90 (10.3)
##   History of COPD = Yes (%)                                61 ( 7.0)
```

```
##   History of Diabetes = Yes (%)                          234 (26.7)
##   History of CHF = Yes (%)                                34 ( 3.9)
##   History of Hypertension = Yes (%)                      395 (45.0)
##   History of Malignancy = Yes (%)                         30 ( 3.4)
##   History of Renal Impairment = Yes (%)                   81 ( 9.2)
##   Immunosuppressed Status = Yes (%)                       65 ( 7.4)
##   Body Mass Index (BMI) (mean (SD))                    32.74 (8.76)
##   NEWS Score at Baseline (mean (SD))                    4.17 (2.07)
##   Baseline Creatinine (mg/dL) (mean (SD))               1.16 (1.67)
##   Baseline C-Reactive Protein (CRP) (mean (SD))        93.28 (69.25)
##   Baseline Lymphocyte Count (mean (SD))                 1.01 (2.83)
##   Pulmonary Ordinal Scale (mean (SD))                   3.18 (0.71)
##   Viral Load Result (Log10) (mean (SD))                 4.21 (1.76)
```

Problem 2

```
## Mean Difference in Hospital-Free Days:

## Treated Group Mean: 19.84607

## Control Group Mean: 20.95758

## Mean Difference (Treated - Control): -1.11151

## 95% Confidence Interval: ( -1.927391 , -0.2956282 )

## z-score: -2.670191

## p-value: 0.007580821
```

Problem 3

This regression model is fit with the existing covariates. In this case, the ATT is the same as the ATE because there are no interaction terms so the predicted response has the same difference between treatment and control for all values.

```
## Average Treatment Effect (ATE):

## - Estimate: 0.3270119

## - Standard Error: 0.4111715

## - 95% Confidence Interval: ( -0.4788842 , 1.132908 )

## - p-value: 0.4265739

## Average Treatment Effect among the Treated (ATT):

## - Estimate: 0.3270119

## - Standard Error: 0.6468967

## - 95% Confidence Interval: ( -0.9409057 , 1.594929 )
```

This regression model includes interaction terms. This was added in order to be able to show a meaningful ATT (not the same as the ATE). Forward stepwise variable selection was used to avoid too many insignificant covariates. Different choices of the penalty for additional covariates in stepwise variable selection were experimented with and these resulted in significantly different models and treatment effects. The model shown uses the standard AIC penalty. The estimate and standard error from the model can't be used as the ATE and standard error of the ATE due to the presence of interaction terms so they are calculated manually.

```
## Average Treatment Effect (ATE):
```

```
## - Estimate: 0.1753556
## - Standard Error: 0.02677391
## - 95% Confidence Interval: ( 0.1228788 , 0.2278325 )
## - p-value: 0.01361855
## Average Treatment Effect among the Treated (ATT):
## - Estimate: -0.05307161
## - Standard Error: 0.03238652
## - 95% Confidence Interval: ( -0.1165492 , 0.01040597 )
```

Problem 4

Part a

```
## Weighted Standardized Mean Differences of all Covariates:

##        gender      asthmahx        copdhx    diabeteshx         chfhx
## -0.0169906487  0.0112755182 -0.0032290677  0.0161410971  0.0001135309
##   hypertenshx      malignhx       renalhx   immuno_supp           age
##  0.0150272580  0.0112291018  0.0244150377  0.0269035489  0.0260192271
##           bmi        newsbl       creatbl         crpbl       lymphbl
## -0.0099095066 -0.0145834930  0.0407633137 -0.0425449051 -0.0142318266
##     ordpulmbl vl_rslt_log10
## -0.0427275289  0.0116400261
```

Part b

```
## ATE using Propensity Score Stratification:
## - Estimate: -0.08095513
## - Standard Error: 0.02861398
## - 95% CI: -0.1370385 -0.02487173
## ATT using Propensity Score Stratification:
## - Estimate: -0.3242481
## - Standard Error: 0.03778354
## - 95% CI: -0.3983039 -0.2501924
```

Part c

```
## ATE using IPW:
## - Estimate: 0.1872562
## - 95% CI: -0.7917661 1.166279
## ATT using IPW:
## - Estimate: -0.397127
## - Standard Error: 0.2703893
## - 95% CI: ( -0.92709 , 0.1328361 )
```

Part d

```
## ATT using Propensity Score Matching:
## - Estimate: 0.04529201
## - Standard Error: 0.3572174
## - 95% Confidence Interval: ( -0.6548541 , 0.7454382 )
```

Problem 5

Looking at each of these estimates, we can see that the unadjusted mean difference reflects a significant negative average treatment effect, but the ATE and ATT estimates when using both outcome regression models, propensity score stratification, inverse probability weighting, and propensity score matching are all much closer to zero with zero being included in most of the 95% confidence intervals for these estimates, indicating that we can not draw much of a conclusion either way regarding the effectiveness of this treatment.